FIT2179 Data Visualisation

Course Notes

AREA CARTOGRAMS

23 September 2026

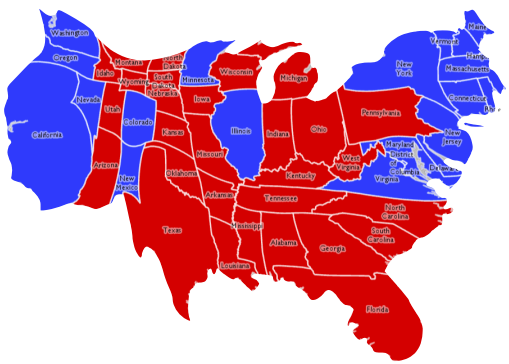
AREA CARTOGRAMS

Area cartograms vary the size of areas so that each area is proportional to a quantitative attribute. This requires an optimisation algorithm that iteratively adjusts the size of the areas, and a variety of such algorithms have been proposed. Colour is often used to show a second quantitative or categorical attribute. Area cartograms are unusual eye-catchers that grab the attention of readers.

Area cartograms differ from traditional maps in that they do not use a map projection. The projection is effectively replaced by an algorithm that distorts space in order to adjust the size of area marks. There are three main types of area cartogram: contiguous, non-contiguous, and Dorling cartograms.

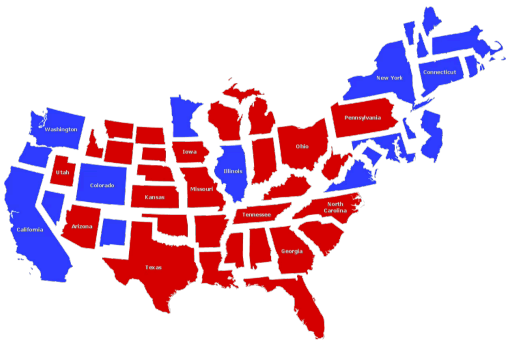
CONTIGUOUS CARTOGRAMS

Contiguous cartograms distort areas without breaking the connections between adjacent areas. Every area keeps its neighbours, but the resulting shapes can be highly distorted. This type has become popular; worldmapper.org holds a large collection of examples.



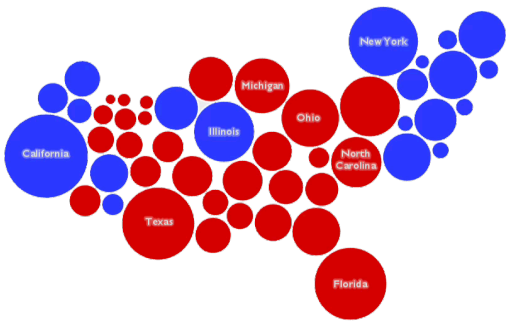
NON-CONTIGUOUS CARTOGRAMS

Non-contiguous cartograms also scale areas in proportion to the attribute value, but the shapes themselves are not otherwise distorted. The scaled areas are arranged so that the original spatial arrangement is retained as far as possible, but they no longer touch one another.



DORLING CARTOGRAMS

Dorling cartograms, named after their inventor, replace each area with a circle whose area is proportional to the mapped value. The circles are arranged so that the neighbourhood relationships between the original areas are retained as far as possible.

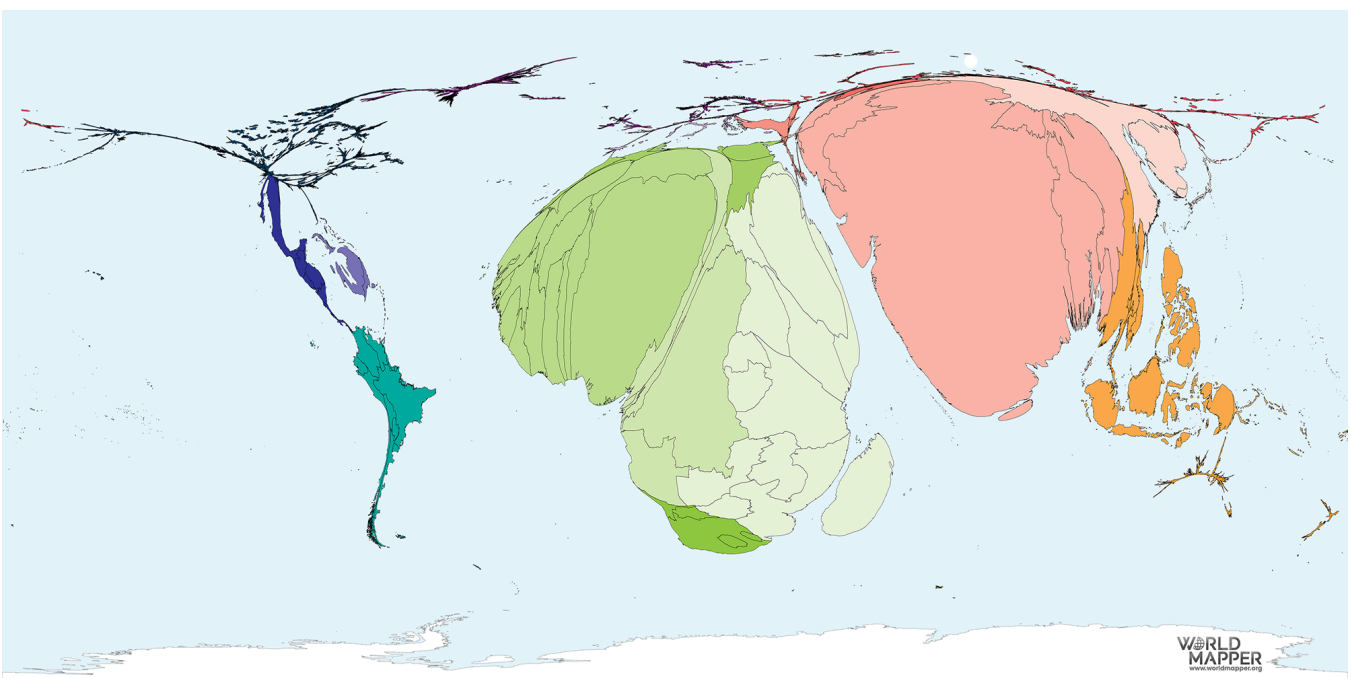


READING CARTOGRAMS

Two difficulties affect area cartograms.

The first is that readers need to be familiar with the size and shape of the undistorted areas. Readers locate places on a map by recognising familiar outlines, but contiguous cartograms can deform outlines beyond recognition in order to encode the attribute. Non-contiguous cartograms keep each shape intact and so remain the most recognisable of the three. Dorling cartograms discard shape entirely, and depend on the reader being familiar with the arrangement of the areas.

The second difficulty is size comparison. The attribute is encoded in area, and readers systematically underestimate how much larger one area is than another. This affects contiguous cartograms most, because areas with irregular outlines are harder to compare than the regular circles of a Dorling cartogram.



Earthquakes Homeless 2000-2025 by World Mapper